

Customer Retention & Churn Analysis Report

1. Introduction

Customer retention is one of the most important factors for business growth and long-term profitability. Retaining existing customers is generally more cost-effective than acquiring new customers. This project focuses on analyzing customer churn behaviour to identify the major reasons why customers leave a service.

The analysis was performed using Python and various data analysis libraries to extract meaningful insights from the customer dataset.

2. Objective of the Project

The primary objectives of this project are:

- To analyze customer churn behaviour
 - To identify factors affecting customer retention
 - To perform exploratory data analysis (EDA)
 - To visualize customer patterns and trends
 - To generate business recommendations for reducing churn
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3. Dataset Description

The dataset used for this project is the Telco Customer Churn Dataset. The dataset contains customer-related information including demographic details, subscription information, account details, and churn status.

Important Features in Dataset

- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Internet Service
- Contract Type
- Payment Method
- Monthly Charges

- Total Charges
- Churn Status

The target variable in the dataset is:

- **Churn** (Yes/No)
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4. Tools and Technologies Used

The following technologies and libraries were used in this project:

Tool / Library	Purpose
Python	Data analysis and processing
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook Development environment	

5. Methodology

5.1 Data Collection

The dataset was imported into Python using the Pandas library.

5.2 Data Cleaning

Data cleaning steps included:

- Handling missing values
- Converting data types
- Checking duplicate records
- Correcting inconsistent values

5.3 Exploratory Data Analysis (EDA)

EDA was performed to understand customer behaviour and identify trends related to churn.

The following analyses were performed:

- Churn distribution analysis
- Gender vs churn analysis
- Contract type analysis
- Internet service analysis

- Monthly charges distribution
- Tenure analysis
- Correlation analysis

5.4 Data Visualization

Different charts and graphs were used to represent the analysis results:

- Count plots
 - Histograms
 - Heatmaps
 - Box plots
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6. Key Findings and Insights

6.1 Contract Type and Churn

Customers with month-to-month contracts showed the highest churn rate compared to customers with long-term contracts.

6.2 Monthly Charges Impact

Customers paying higher monthly charges were more likely to leave the service.

6.3 Customer Tenure

Customers with lower tenure had a significantly higher probability of churn. Long-term customers were generally more loyal.

6.4 Internet Service

Fiber optic internet users experienced higher churn compared to DSL users.

6.5 Payment Method

Customers using electronic check payment methods had higher churn rates than customers using automatic payment methods.

7. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Provide discounts for long-term subscription plans
- Improve customer onboarding experience
- Enhance customer support quality
- Introduce loyalty and reward programs
- Improve service quality for high-risk customer groups
- Encourage customers to use automatic payment methods

8. Conclusion

The Customer Retention and Churn Analysis project successfully identified important factors influencing customer churn. The analysis revealed that contract type, tenure, monthly charges, internet service, and payment methods significantly impact customer retention.

Using data analysis and visualization techniques, meaningful business insights were generated that can help organizations improve customer satisfaction and reduce churn rates.

This project demonstrates practical applications of Python in business analytics, customer behavior analysis, and data-driven decision-making.

9. Future Scope

Future improvements for this project may include:

- Building machine learning models for churn prediction
 - Developing interactive dashboards using Streamlit or Power BI
 - Performing customer segmentation analysis
 - Implementing real-time churn monitoring systems
 - Applying advanced predictive analytics techniques
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10. References

- Telco Customer Churn Dataset
- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation